

Literature Answer to Problem 1.1 of arXiv:1910.02414

FA/Banach run packet

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Status

This is a literature-already-answered packet. It records that Problem 1.1 from Albiac–Ansorena–Dilworth–Kutzarova [1] was answered affirmatively in the later paper of Casazza–Dilworth–Kutzarova–Motakis [2]. It is not an original proof from this run.

Original Problem Evidence

The original problem appears as Problem 1.1 on page 4 of [1].

- (b) there is (up to equivalence) a unique symmetric basic sequence;
- (c) there exist finitely many non-equivalent symmetric basic sequences; and
- (d) there are infinitely many nonequivalent symmetric basic sequences.

The case when a Banach space has a unique symmetric basic sequence deserves special attention. Note that amongst all the aforementioned examples, only ℓ_p and c_0 have a unique symmetric basic sequence. Altshuler constructed in [6] a Banach space with a symmetric basis which does not contain any subspace isomorphic to ℓ_p or c_0 , and whose symmetric basic sequences are all equivalent. Another remarkable example was provided by Read [24]. He answered a question of Lindenstrauss and Tzafriri by constructing a space with exactly two symmetric bases (up to equivalence). More precisely, Read proved that every symmetric basic sequence is equivalent either to the unit vector basis of ℓ_1 or to one of the two symmetric bases of his space. It was remarked in [17] that a careful look at the papers of Altshuler and Read shows that their proofs work similarly for the more general case of all subsymmetric basic sequences. Let us restate some of the questions raised in [17] about the richness of subsymmetric sequences.

Problem 1.1. Does there exist a Banach space in which all subsymmetric basic sequences are equivalent to one basis, and that basis is not symmetric?

Problem 1.2. Does there exist a Banach space with, up to equivalence, precisely two subsymmetric bases, at least one of them is not symmetric?

In addition, we may also ask the following.

Problem 1.3. Does there exist a Banach space with, up to equivalence, precisely n subsymmetric basic sequences such that at least one of them is not symmetric? Here n is a natural number greater than 1.

With an eye on this kind of questions, in Section 2 we develop new tools which lead to a dichotomy theorem for general Banach spaces with a subsymmetric basis. We prove that every subsymmetric sequence

In words, the problem asks whether there is a Banach space in which all subsymmetric basic sequences are equivalent to a single basis, while that basis is not symmetric.

Supporting Answer Evidence

The later paper [2] explicitly cites the question from [3, 1] and says that it answers it positively.

played an important role by themselves within the general theory. The first non-arbitrarily distortable space constructed by Schlumprecht [22] has a subsymmetric basis.

Altshuler [3] (see also [19]) constructed a space in which all symmetric basic sequences are equivalent to its symmetric basis. Later, another example of this kind was built in [8], based on a symmetric version $S(T^*)$ of T^* . Recently, Albiac, Ansorena and Wallis [4] used Garling-type spaces to provide the first example of a Banach space with a unique subsymmetric basis which is not symmetric. However, as shown in a sequel paper [5], that space contains a continuum of non-equivalent subsymmetric basic sequences. A careful inspection of Altshuler's proof shows that, in fact, all subsymmetric basic sequences of his space are also equivalent to the symmetric basis. The same turned out to be true in the case of $S(T^*)$.

In view of Altshuler's example, it was asked in [18] and [5] whether there exists a space with a subsymmetric basis with a unique, up to equivalence, subsymmetric basic sequence which is not symmetric. In this paper we answer this question in the positive. For this purpose, we use the subsymmetric version $Su(T^*)$ [8] of the original Tsirelson space T^* [23]. One can also define a subsymmetric version of Altshuler's space, however we do not know whether all its subsymmetric basic sequences are equivalent. In addition, we show that the dual space $Su(T^*)^*$ has no symmetric basic sequences. Next, we extend to subsymmetric bases Altshuler's characterization [2] of the unit vector bases of c_0 and ℓ_p among symmetric bases and we give various sufficient conditions for a space with a subsymmetric (resp. symmetric) basis to have a continuum of subsymmetric (resp. symmetric) basic sequences.

Idea of the Answer

The answer uses the subsymmetrization $Su(T^*)$ of Tsirelson's original space. The canonical basis of $Su(T^*)$ is built to be subsymmetric but not symmetric. The main rigidity result in the supporting paper shows that every subsymmetric basic sequence in $Su(T^*)$ is equivalent to this canonical basis. Thus the basis supplies the required unique equivalence class, and its failure of symmetry supplies the required non-symmetric witness.

Literature Answer. *Problem 1.1 of [1] has an affirmative answer. The Banach space $Su(T^*)$ from [2] has, up to equivalence, a unique subsymmetric basic sequence, and its canonical subsymmetric basis is not symmetric.*

Proof. Problem 1.1 in [1] asks for a Banach space whose subsymmetric basic sequences are all equivalent to one basis, with that basis non-symmetric.

The later paper [2] identifies this question in its introduction and states that it answers the question positively. The witness is the subsymmetrization $Su(T^*)$ of Tsirelson's original space. In Section 2 of [2], the authors state that the canonical basis of $Su(T^*)$ is not symmetric, prove that the norm of $Su(T^*)$ is 1-subsymmetric, and prove that every subsymmetric sequence in $Su(T^*)$ is equivalent to the canonical basis. These statements give exactly the existence requested in Problem 1.1. $\square$

Limitations

This packet answers only Problem 1.1 from [1]. The adjacent Problems 1.2 and 1.3 ask for exactly two, or exactly $n > 1$, subsymmetric basic sequences with at least one non-symmetric class. The supporting paper is not used here as an answer to those finite-number questions.

Verification Notes

The source and supporting papers are distinct. The supporting paper explicitly mentions the question from [1], so this belongs in `literature_already_answered` rather than `literature_implied_answers`. The local duplicate search covered arXiv ids 1910.02414 and 2112.09602, plus the phrases `unique subsymmetric`, `non-symmetric`, `Su(T*)`, and `Tsirelson`.

References

- [1] F. Albiac, J. L. Ansorena, S. J. Dilworth, and D. Kutzarova, *A dichotomy for subsymmetric basic sequences with applications to Garling spaces*, Trans. Amer. Math. Soc. 374 (2021), no. 3, 2079–2106; arXiv:1910.02414.
- [2] P. G. Casazza, S. J. Dilworth, D. Kutzarova, and P. Motakis, *On uniqueness and plentitude of subsymmetric sequences*, arXiv:2112.09602.
- [3] D. Kutzarova, B. S. Mityagin, and K. Papathanasiou, *On subsymmetric basic sequences*, Studia Math. 218 (2013), no. 1, 1–21.